

The Pasting Lemma (§1.62)

by

§1.62 — the intersection $\rightarrow$ union square of two subobjects is a pushout

In a pre-logos, for subobjects $A_1, A_2 \hookrightarrow A$ with intersection $A_1 \cap A_2$ and union $A_1 \cup A_2$, this square of monics is a **pushout**:

$$\begin{array}{ccc} A_1 \cap A_2 & \xrightarrow{\bar{x}} & A_1 \\ \bar{y} \downarrow & & \downarrow x \\ A_2 & \xrightarrow{y} & A_1 \cup A_2 \end{array}$$

Pushout means: a function out of A_1 and one out of A_2 that **agree on the overlap** glue to a unique function on $A_1 \cup A_2$. In SET with $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 3, 4\}$, overlap $A_1 \cap A_2 = \{2, 3\}$, union $A_1 \cup A_2 = \{1, 2, 3, 4\}$: given f on A_1 and g on A_2 with $f = g$ on $\{2, 3\}$, the glued map sends 1 by f , 4 by g , and 2, 3 by either (they agree).

§1. Why it is a pushout: $R = x^\circ f \cup y^\circ g$ is a map

A relation is a **map** iff $1 \subseteq RR^\circ$ (**entire**) and $R^\circ R \subseteq 1$ (**simple**) — §1.564. Given $f : A_1 \rightarrow Q$, $g : A_2 \rightarrow Q$ agreeing on the overlap ($\bar{x}f = \bar{y}g$), set $R = x^\circ f \cup y^\circ g : A \rightarrow Q$. Four relational facts close it (diagram order; checked in SET):

- | | |
|---|---|
| 1. $xx^\circ = 1$ | — x is monic |
| 2. $1 \subseteq ff^\circ$ | — f is a map (entire) |
| 3. $x^\circ x \cup y^\circ y = 1$ | — x, y cover A (pre-logos: the two image-coreflexives fill A) |
| 4. $(\bar{x}f)^\circ(\bar{x}f) \subseteq 1$ | — $\bar{x}f$ is a map (simple) |

Entire. By (3) then (2): $1 = x^\circ x \cup y^\circ y \subseteq x^\circ ff^\circ x \cup y^\circ gg^\circ y \subseteq RR^\circ$.

Simple. $R^\circ R$ expands to four terms; the diagonal ones reduce by (1) to $f^\circ f, g^\circ g \subseteq 1$, and the cross terms reduce — via the tabulation $xy^\circ = \bar{x}^\circ \bar{y}$ of the intersection and the agreement $\bar{x}f = \bar{y}g$ — to $(\bar{x}f)^\circ(\bar{x}f) \subseteq 1$ by (4). So $R^\circ R \subseteq 1$.

Hence R is a map. Then $xR = xx^\circ f \cup xy^\circ g = f$ (by (1), the second term absorbed), similarly $yR = g$, and R is **unique** because x, y cover.

★ Punchline

The mediating map is forced by the relational calculus: $R = x^\circ f \cup y^\circ g$ is entire because the inclusions cover, simple because everything in sight is a map, and unique because the inclusions are jointly epic. The one pre-logos-only input is that **union distributes over composition** (§1.616) — which is why the lemma fails in a bare regular category.

§2. §1.565 vs §1.62 — same method, different square

Both build a mediating map as a union/intersection of relations and check it is a map. They differ in **which** arrows and **which** category:

§1.62 — monics, pre-logos

$$\begin{array}{ccc}
 A_1 \cap A_2 & \rightharpoonup & A_1 \\
 \downarrow & & \downarrow \\
 A_2 & \rightarrow & A_1 \cup A_2
 \end{array}$$

Mediator $R = x^\circ f \cup y^\circ g$. Needs §1.616 (union over composition).

§1.565 — covers, regular

$$\begin{array}{ccc}
 P & \twoheadrightarrow & B \\
 \downarrow & & \downarrow \\
 C & \twoheadrightarrow & D
 \end{array}$$

Mediator $R = (x^\circ u) \cap (y^\circ u)$. Holds in any regular category.

One glues monics (subobjects) into their union; the other glues a pullback of covers (epis). Dual sides of the factorization — and only the monic side needs the extra pre-logos axiom.